They must show the tests clearly and use absolute value when needed.

MATH 152 **QUIZ 8A** NAME Solution

Mrs. Bonny Tighe 25 points

11.8 TA_____ Wed 11/7/2018

5 pts

1. Find the radius of convergence and interval of convergence of the power series.

a) $\sum_{n=1}^{\infty} \frac{(1\cdot 3\cdot 5\cdots(2n-1))(3x-1)^n}{5^{n+1}}$

$$\lim_{n\to\infty}\left|\frac{\cancel{1\cdot 3\cdot 5\cdots(2n-1)}(2n+1)\cancel{(3x-1)(3x-1)^n}}{\cancel{5\cdot 5^{n+1}}}\cdot\frac{\cancel{5^{n+1}}}{\cancel{1\cdot 3\cdot 5\cdots(2n-1)}\cancel{(3x-1)^n}}\right|$$

$$\lim_{n\to\infty}\left|\frac{3x-1}{5}\cdot(2n+1)\right| = \infty \text{ numbers from } 1$$

so $R = 0$

$I = \{1/3\}$

5 pts

b) $\sum_{k=0}^{\infty} \frac{(x-3)^{2k}}{6^k}$ Geometric Series Test

$r = \left|\frac{(x-3)^2}{6}\right| < 1$ $\quad |(x-3)^2| < 6$ $\quad |x-3| < \sqrt{6}$

$R = \sqrt{6}$

$-\sqrt{6} < x-3 < \sqrt{6}$

$+3-\sqrt{6} < x < 3+\sqrt{6}$

$I = (3-\sqrt{6},\ 3+\sqrt{6})$

do not need to test end points

2 pts

2. Expand $\sum_{n=0}^{\infty} \frac{(-1)^n(x-1)^n}{4^{n+1}(n+2)}$ as a polynomial to the third degree.

$$= \frac{1}{8} - \frac{(x-1)}{4^2(3)} + \frac{(x-1)^2}{4^3(4)} - \frac{(x-1)^3}{4^4(5)}$$

3. Find the radius of convergence and interval of convergence of the power series.

5pts

a) $\sum_{n=1}^{\infty}(-1)^n \frac{(x+1)^n}{\sqrt{n+3}}$

$\lim_{n\to\infty}\left|\frac{(x+1)(x+1)^n}{\sqrt{n+4}}\cdot\frac{\sqrt{n+3}}{(x+1)^n}\right| =$

$\lim_{n\to\infty}\left|(x+1)\frac{\sqrt{n+3}}{\sqrt{n+4}}\right| = |x+1| < 1$

$-1 < x+1 < 1$

$-2 < x < 0$

$R = 1$

$I = (-2, 0]$

Test end points

$x = -2$

$\sum_{n=1}^{\infty}\frac{(-1)^n(-1)^n}{\sqrt{n+3}}$

$\sum_{n=1}^{\infty}\frac{1}{\sqrt{n+3}}$ divergent

$x = 0$

$\sum_{n=1}^{\infty}\frac{(-1)^n 1^n}{\sqrt{n+3}}$

$\sum_{n=1}^{\infty}\frac{(-1)^n}{\sqrt{n+3}}$ conditionally converges

5pts

b) $\sum_{n=1}^{\infty}\frac{(-1)^n\ 5^n\ (x+2)^n}{1\cdot 4\cdot 7\cdots(3n-2)}$

$\lim_{n\to\infty}\left|\frac{5\cdot 5^n\ (x+2)\cdot(x+2)^n}{1\cdot 4\cdot 7\cdots(3n-2)(3n+1)}\cdot\frac{1\cdot 4\cdot 7\cdots(3n-2)}{5^n\ (x+2)^n}\right|$

$\lim_{n\to\infty}\left|5(x+2)\cdot\frac{1}{3n+1}\right| = 0$ always less than 1

$R = \infty$

$I = (-\infty, \infty)$

3pts

4. If $\sum_{n=0}^{\infty}c_n x^n$ is convergent when $x = -5$ and divergent when $x = 8$, decide which of the following are convergent or divergent. a) $\sum_{n=0}^{\infty}c_n(-3)^n$ Convergent

b) $\sum_{n=0}^{\infty}c_n$ Convergent c) $\sum_{n=0}^{\infty}c_n 9^n$ Divergent

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5pts 1. Find the radius of convergence and interval of convergence of the power series.

a) $\sum_{n=1}^{\infty} \frac{(-1)^n\ 5^n\ (x+2)^n}{1\cdot 3\cdot 5\cdots(2n-1)}$

$$\lim_{n\to\infty}\left|\frac{5\cdot 5^n (x+2)(x+2)^n}{1\cdot 3\cdot 5\cdots(2n-1)(2n+1)}\cdot\frac{1\cdot 3\cdot 5\cdots(2n-1)}{5^n(x+2)^n}\right|$$

$$\lim_{n\to\infty}\left|5(x+2)\cdot\left(\frac{1}{2n+1}\right)\right| = 0 < 1 \text{ always}$$

$R = \infty$

$I = (-\infty, \infty)$

5pts b) $\sum_{k=0}^{\infty} \frac{(x+2)^{2k}}{5^{2k}}$ Can use geometric series test

$r = \left|\frac{x+2}{5}\right| < 1$

$|x+2| < 5$

$-5 < x+2 < 5$

$-7 < x < 3$

$R = 5$

$I = (-7, 3)$

Do not need to test endpoints of Geometric series test was used

2pts 2. Expand $\sum_{n=0}^{\infty} \frac{(-1)^n (x+1)^n}{3^{n+1} n!}$ as a polynomial to the third degree.

$$= \frac{1}{3} - \frac{(x+1)}{3^2 1!} + \frac{(x+1)^2}{3^3(2!)} - \frac{(x+1)^3}{3^4(3!)}$$

3. Find the radius of convergence and interval of convergence of the power series.

5pts a) $\sum_{n=1}^{\infty}(-1)^n \frac{(x-2)^n}{n\sqrt{n+3}}$

$$\lim_{n\to\infty}\left|\frac{(x-2)\cancel{(x-2)^n}}{(n+1)\sqrt{n+4}}\cdot\frac{n\sqrt{n+3}}{\cancel{(x-2)^n}}\right| = \lim_{n\to\infty}\left|(x-2)\left(\frac{n\sqrt{n+3}}{(n+1)\sqrt{n+4}}\right)\right|$$

$= |x-2| < 1$

$-1 < x-2 < 1$

$1 < x < 3$

$R = 1$

$I = [1, 3]$

Must test end points

$x = 1$

$$\sum_{n=1}^{\infty}\frac{(-1)^n(-1)^n}{n\sqrt{n+3}}$$

$$\sum_{n=1}^{\infty}\frac{1}{n\sqrt{n+3}} \text{ convergent}$$

$x = 3$

$$\sum_{n=1}^{\infty}\frac{(-1)^n 1^n}{n\sqrt{n+3}}$$

$$\sum_{n=1}^{\infty}\frac{(-1)^n}{n\sqrt{n+3}} \text{ absolutely convergent}$$

5pts b) $\sum_{n=1}^{\infty}\frac{(1\cdot 4\cdot 7\cdots(3n-2))(x+2)^n}{5^n}$

$$\lim_{n\to\infty}\left|\frac{1\cdot 4\cdot 7\cdots(3n-2)(3n+1)(x+2)\cancel{(x+2)^n}}{5\cdot\cancel{5^n}}\cdot\frac{\cancel{5^n}}{1\cdot 4\cdot 7\cdots(3n-2)\cancel{(x+2)^n}}\right|$$

$$\lim_{n\to\infty}\left|\frac{x+2}{5}(3n+1)\right| = \infty \text{ never less than } 1$$

so $R = 0$

$I = \{-2\}$

3pts 4. If $\sum_{n=0}^{\infty}c_n x^n$ is convergent when $x = -5$ and divergent when $x = 8$, decide which of the following are convergent or divergent. a) $\sum_{n=0}^{\infty}c_n(-3)^n$ Convergent

b) $\sum_{n=0}^{\infty}c_n$ Convergent c) $\sum_{n=0}^{\infty}c_n 9^n$ Divergent